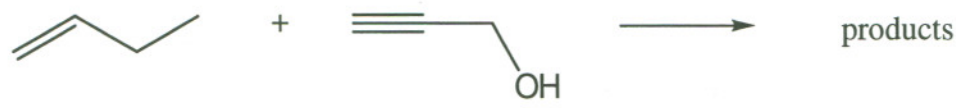
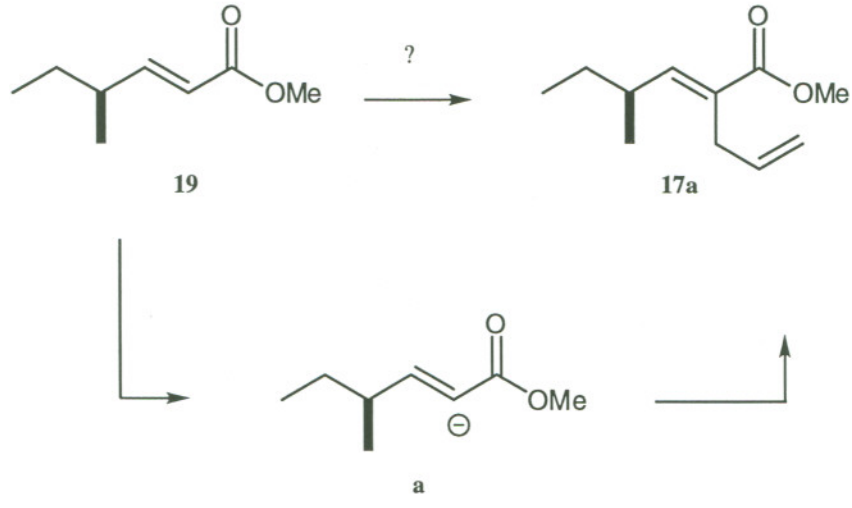
Organic Division Cumulative Exam October 3, 1998

Reference: Trost, Probst & Schoop, *J. Am. Chem. Soc.* **1998**, *120*, 9228-9236

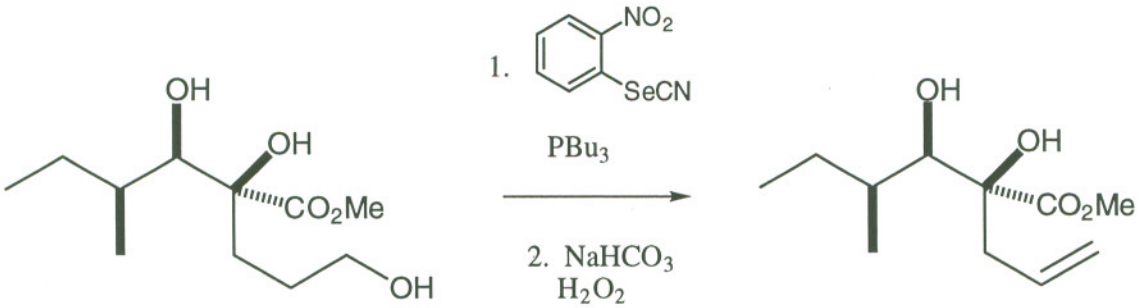
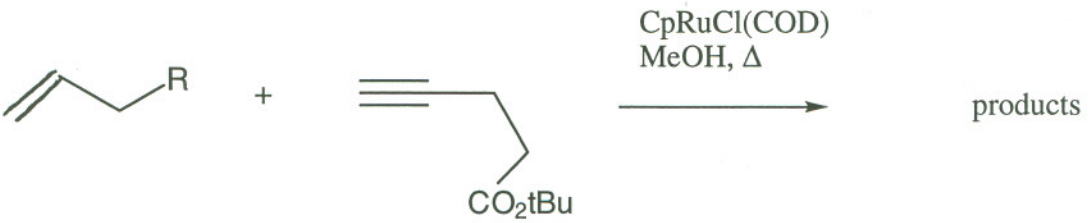
1. Using electron pushing arrows, show how two isomeric Alder ene products could be formed in the <u>thermal</u> Alder ene reaction.



2. Enoate **19** could formally be converted to diene **17a** by deprotonation of **19** by strong base to form anion **a**, followed by alkylation. Why would this approach <u>not</u> work (i.e. what would be the likely side product(s) of such a reaction), and how was the conversion of **19** to **17a** actually achieved (include reagents)?



3. What is the mechanism of the following reaction? What is the general name of the alkene-forming step?



4. Draw the mechanism of the following reaction and suggest an explanation as to why only two of the four possible products are observed.